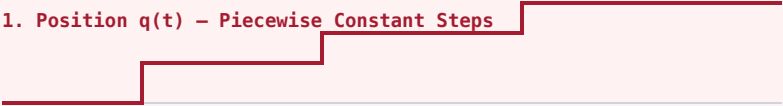


DERIVATIVE CONTINUITY (C^2) VS. DISCONTINUOUS WAYPOINT STEPPING

How quintic polynomial spline interpolation bounds acceleration and eliminates infinite jerk spikes ($\delta(t)$)

✗ DISCONTINUOUS C0 STEPPING (ZERO-ORDER HOLD)

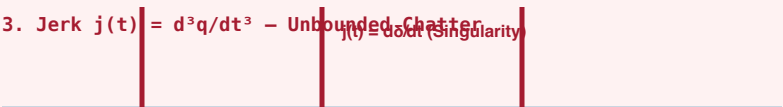
1. Position $q(t)$ – Piecewise Constant Steps



2. Velocity $\dot{q}(t)$ – Infinite Dirac Spikes $\delta(t)$
 $\dot{q} \rightarrow \infty$



3. Jerk $j(t) = d^3q/dt^3$ – Unbounded Chatter
 $j(t) = \text{infinite (singularity)}$



PHYSICAL HARDWARE CONSEQUENCES:

- **Extreme Torque Spikes:** Motor commanded to accelerate instantly.
- **Current Inrush ($I^2 R$):** Massive coil heating, blowing power fuses.
- **Gearbox Backlash & Wear:** Strips reduction gear teeth.
- **Audio Acoustics:** Loud, high-pitched mechanical knocking.

✓ C2 CONTINUITY (QUINTIC POLYNOMIAL SPLINE)

1. Position $q(t)$ – Smooth Quintic Interpolation



2. Velocity $\dot{q}(t)$ – Continuous & Bounded ($|\dot{q}| \leq v_{\text{max}}$)



3. Jerk $j(t) = d^3q/dt^3$ – Bounded & Smooth



PHYSICAL SYSTEMS ADVANTAGES:

- **Smooth Motor Commutation:** Current follows gradual sinusoidal curves.
- **Thermal Protection:** Operates within nominal continuous torque ratings.
- **10x Extended Lifespan:** Zero harmonic resonance or gearbox damage.
- **Rock-Solid Tracking:** 1 kHz MCU PID loops track trajectory perfectly.